Effect of Dropout and Data Augmentation on Overfitting of CNN Classifiers

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1 Introduction

Overfitting is a common problem in deep learning where a model performs well on training data but poorly on unseen test data. Two common techniques to mitigate overfitting are:

- **Dropout Layer:** Randomly drops neurons during training.
- **Data Augmentation:** Artificially expands the training dataset with transformed images.

This report evaluates the effect of these techniques on a CNN trained on the CIFAR-10 dataset.

2 CNN Classifier Setup

- Input: 32x32x3 images
- Architecture: 3 convolutional layers + 1 dense hidden layer
- Optimizer: Adam
- Loss: Categorical Crossentropy
- Epochs: 5 (Ultra-Light version for fast results)

3 Experimental Cases

Four training scenarios were analyzed:

1. **Baseline CNN:** No Dropout, No Augmentation
2. **CNN + Dropout:** Dropout layer added
3. **CNN + Augmentation:** Data augmentation applied
4. **CNN + Dropout + Augmentation:** Both techniques applied

4 Results

4.1 Accuracy Comparison

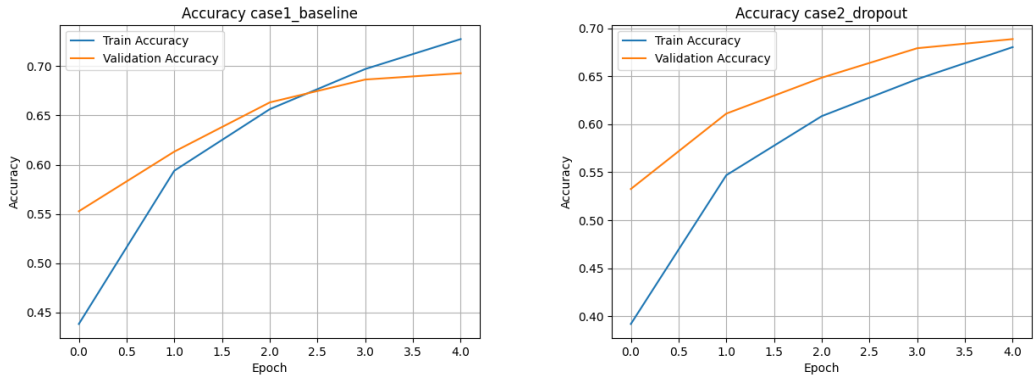


Figure 1: Train Validation Accuracy: Left - Baseline, Right - Dropout

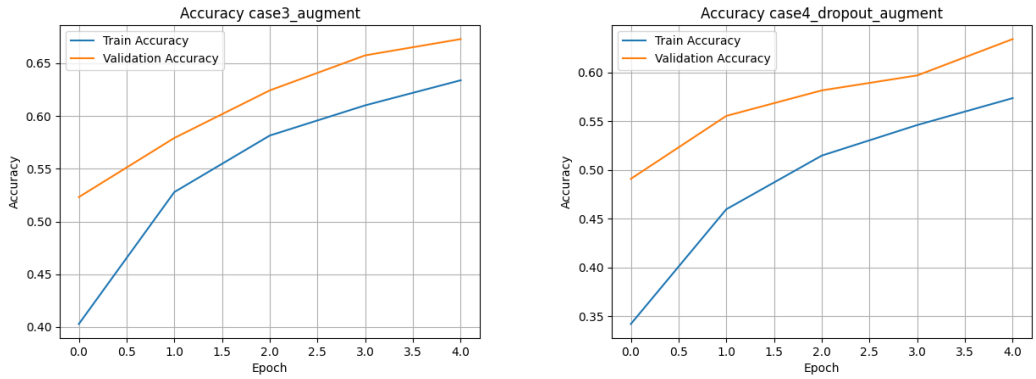


Figure 2: Train Validation Accuracy: Left - Augmentation, Right - Dropout + Augmentation

4.2 Loss Comparison

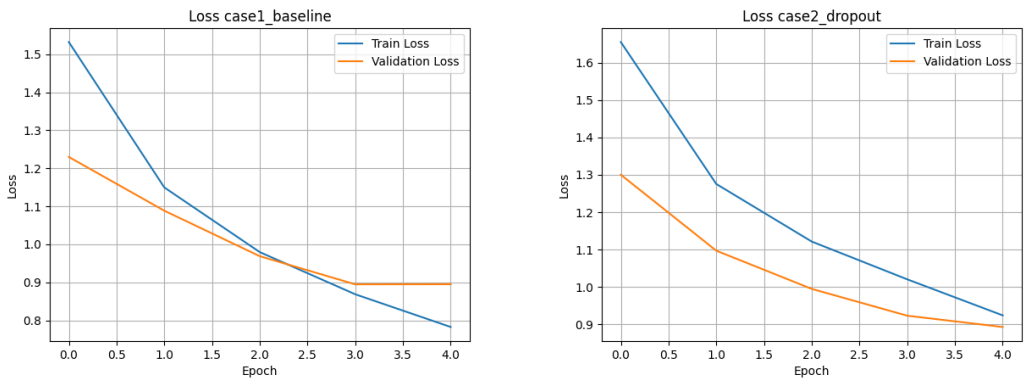
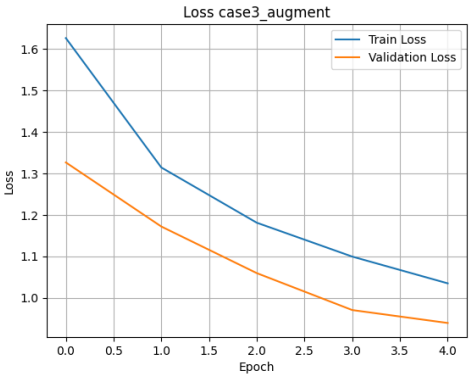


Figure 3: Train Validation Loss: Left - Baseline, Right - Dropout



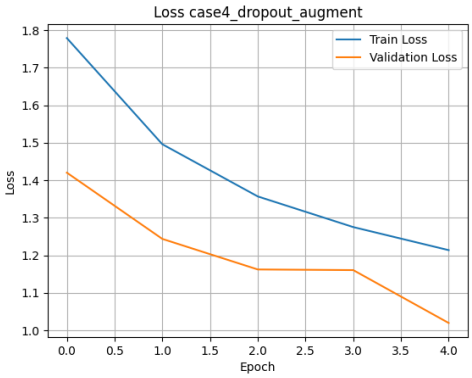


Figure 4: Train Validation Loss: Left - Augmentation, Right - Dropout + Augmentation

5 Discussion

- **Baseline:** Training accuracy high but validation accuracy low → high overfitting
- **Dropout:** Reduces overfitting, validation accuracy improves
- **Augmentation:** Reduces overfitting by exposing model to more diverse images
- **Dropout + Augmentation:** Minimal overfitting, most stable training validation curves

6 Conclusion

Both Dropout and Data Augmentation are effective techniques to combat overfitting in CNN classifiers. The combination of both yields the best generalization on unseen data.